



University
of Exeter

COURSEWORK SPECIFICATION

ECM2419 – Database Theory and Design

Module Leader: **Dr Zeliang Wang**

Academic Year: 2026/27

Title: **Coursework for ECM2419 Database Theory and Design**

Submission deadline: **12 noon, 10th November 2026**

This assessment contributes **40%** of the total module mark and assesses the following **intended learning outcomes**:

- Demonstrate an understanding of relational theory and its use in database design;
- Interpret an informal requirement specification;
- Express designs at different levels of abstraction and map between them;
- Systematically analyse information and make appropriate design choices;
- Write a technical report incorporating the design, implementation and testing of a database system.

Plagiarism

This is an individual assessment.

Plagiarism is interpreted by the university as the act of presenting the work of others as one's own work, without acknowledgement. It is considered academically fraudulent and an offence against university discipline. Your attention is drawn to the [university's regulations on plagiarism](#).

Generative AI

This assessment falls under the category of **AI-Minimal** in the [University's Guidance on use of GenAI in Assessment](#).

This means: You may use AI tools for checking spelling and grammar mistakes only, with no other impact on the structure or content of the assessment. This is because using GenAI tools outside of these uses prevents fair assessment of your ability to achieve module learning outcomes.

When *writing* your assessment, you must never use AI tools:

- For uses other than checking your spelling and grammar.
- To translate more than a word or short phrase into English.
- To upload sensitive or identifying material to an AI tool
- To present material that has been generated by AI as your own work or the work of someone else.

When *submitting* your assessment, you must:

- Check the box during the submission process, that confirms you have adhered to the university's academic conduct policy and the expectations on use of GenAI in your assessment brief.

Instructions

Introduction

This coursework comprises three main tasks:

Task 1 involves specification analysis, and high-level conceptual database design for a bank account management system. Specifically, you will

- Design an Entity Relationship Diagram (ERD) **using UML notation**. The ERD should include all the necessary entities (including their attributes), multiplicity constraints and relationships needed to meet fully the specifications of the database. Where appropriate you should make use of enhanced ERD.
- Explain and justify systematically all elements of your designed ERD. You should describe and justify the entities (including entity attributes), multiplicity constraints and relationships between the entities that fulfil the specifications of the database. Where appropriate you should make reasonable assumptions to filling relevant missing details.

Task 2 is about normalising a provided data table into 3rd normal form (3NF). Specifically, you will

- Produce normalised relational schemas and instances in different normal forms (i.e., 1NF, 2NF, and 3NF), and identify any partial and transitive dependencies where applicable.
- Explain the process by which you generated them from the data given and justify any assumptions you have made.

Task 3 is about relational algebra.

Task 1 - Conceptual Database Design

Southwest Bank is a financial services provider offering a range of bank accounts to its customers. These include mortgage accounts, savings accounts, current accounts, and credit card accounts. Customers may also hold joint accounts (for example, between a husband and wife). Each type of account has a specific interest rate, a balance, and a credit or overdraft limit, and may incur a monthly fee. Every account is uniquely identified by an account number.

A customer may open an account at any of the provider's branches. Each customer is assigned a specific employee as their personal banker. The personal banker can carry out operations for both new and existing customers, including:

- Opening a new customer account
- Providing services such as withdrawals, deposits, transfers and deleting the customer record when closing an account
- Assigning credit limits to a customer
- Freezing a customer account for security reasons, such as a lost or stolen credit card.

An existing customer can perform operations such as:

- Viewing account details, such as account balances and statements
- Requesting a change of address
- Withdrawing cash
- Depositing cash
- Transferring money into other accounts

The system also keeps a record of all transactions performed on each account. A transaction consists of information such as:

- Transaction type, e.g., withdrawal or deposit
- Transaction date and time
- The account associated with the transaction

Your task is to design and produce an ERD using UML notation based on the specification above. The system will be operated by Southwest Bank, which has numerous branches across the UK. Each branch has an address and a branch number, is headed by a bank manager, and employs several other staff members.

Please note that this is a design task, and you may need to resolve ambiguities, infer or add relevant missing details, and interpret the specification carefully. You are expected to do this autonomously and to explicitly explain and justify all your design decisions.

Task 2 - Database Normalisation

From the module ELE web page (<https://ele.exeter.ac.uk/course/view.php?id=30532>), download the spreadsheet titled ***Normalisation task data table.xlsx***. It can be found in the tile ***Assessment information and submission***, under the section ***Coursework brief***. The spreadsheet contains a sample of bank account transactions for a small firm, with each row representing a unique combination of bank account and transaction date.

Your task is to normalise the provided data table into 3NF. In your solution, you should show all intermediate steps, including the resulting relations (schemas and instances) from 1NF through to 3NF, clearly labelling any partial and transitive dependencies where applicable. You should also justify any assumptions you make.

Task 3 - Relational Algebra

The four relations shown in Figure 1 represent a set of tables in a database which store information about student assessments at Exmouth University. Based on the relations shown in Figure 1:

Table name: STUDENT_1

STU_NUM	STU_LNAME	CRS_CODE
101452	Williams	Comp-101
104257	Smith	Comp-310
104258	Lewis	Comp-101

Table name: STUDENT_2

STU_NUM	STU_LNAME	CRS_CODE
104258	Lewis	Comp-101
104787	Bond	Comp-101

Table name: MODULE

MODULE_CODE	MODULE_NAME
ECM2001	Databases
ECM2012	Computer Network
ECM1010	Programming

Table name: ASSESSMENT

STU_NUM	MODULE_CODE	EXAM_MARK	COURSEWORK_MARK
101452	ECM2001	60	70
101452	ECM1010	50	60
104258	ECM1010	65	65
104457	ECM2012	0	70

Figure 1. Tables from student assessment database

- 1) Write a relational algebraic expression to find the names of all students who are enrolled in 'Comp-101'.
- 2) Write a relational algebraic expression to list the names of all students in STUDENT_1 who scored less than 60 in the 'Programming' exam.
- 3) Write a relational algebraic expression to list the names and student numbers of all students in STUDENT_1 who have achieved 60 or over in both coursework and exam marks.
- 4) Determine the resulting relation produced by the following relational algebra expression. Clearly show all intermediate steps and justify your answer.

$\Pi_{\text{module_name}} (\text{MODULE} \bowtie ((\text{ASSESSMENT}) \bowtie (\sigma_{\text{stu_lname} = \text{'Williams'}} (\text{STUDENT_1}))))$

What You Should Submit

You should submit a **single PDF** document via ELE by **12 noon, Tuesday 10th November 2026**, and your document **must not exceed 8 sides of A4** (excluding the reference page).

Your report must conform to the Department of Computer Science report template. Compliance with the prescribed template is a requirement of this assessment, and failure to comply may result in marks being deducted.

The report will be marked anonymously. You must not include your name or student number on the title page or anywhere else in the report.

There is no single required structure for the report. One possible approach is to organise the report into three separate sections, one for each task. However, you may adapt the structure as appropriate, provided that the report remains logical, coherent, and professional.

Marking Criteria

Task No.	Task Description	Key Aspects to be Assessed	Mark Allocated
1	ERD design of the database	Marks will be awarded for identifying and appropriately naming all the entities , relationships and attributes that ensure that the resulting database design will comply fully with the specifications mentioned in section 2 and any other requirements from your reasonable assumptions. Marks will be awarded for indicating the cardinality of each entity in each relationship, for identifying and labelling the primary keys of all the entities, and for appropriate use of enhanced constructs (e.g., generalisation/specialisation).	30%
	ERD documentation	Marks will be awarded for a systematic description and clear justification of all elements of your design, stating explicitly and justifying any assumptions made. This includes clear justification of your use of entities, relationships and attributes; justification of the choice of the primary keys of all the entities in your database design as well as the cardinality of each entity in a relationship; justification of enhanced construct.	10%
2	Normalisation	Marks will be awarded for the correct relations normalised to 1NF, 2NF and 3NF for the provided data.	20%
	Normalisation documentation	Marks will be awarded for clearly explaining the process for normalisation to 3NF of the provided data showing all intermediate steps, and any assumptions you have made.	10%
3	Relational algebra	Marks will be awarded for the correct answer to for each query.	20%
	Writing and presentation	Marks will be awarded for a writing style that is clear and concise, a well-structured report, and the inclusion of references where applicable.	10%
Penalty of overlength submissions (-10 marks) Penalty of drawing ERD without using UML notation (-10 marks)			